

Appendix C - Abrolhos Marine Park Western Rock Lobster Data Analysis Report

North-west Marine Parks Network

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Data Analysis

Statistical methods and test results underpinning the western rock lobster natural values reported in Appendix A.

Lobster per pot was modelled for each metric with a negative binomial generalised linear mixed model (log link) in `glmmTMB`:

```
count ~ status * year + depth + (1 | string)
```

where `status` is No-Take (National Park Zone) or Fished (Special Purpose Zone), `year` is the survey year, `depth` is standardised pot depth and `string` is the line of pots each pot was set in. The same strings were fished in each year. Pots that caught no lobster are included as zeros.

Status by year interaction

Table C.1: Likelihood ratio test of the status by year interaction for each western rock lobster metric, comparing the full model with a model without the interaction. A significant result indicates the difference between No-Take and Fished changed between survey years.

Metric	Pots	Chi squared	df	p
All lobster	284	0.09	1	0.763
Sub-maturity (<75 mm CL)	284	0.11	1	0.745
Female 75 mm CL	284	0.00	1	0.993
Male 75 mm CL	284	0.75	1	0.385
Female 100 mm CL	284	0.04	1	0.839
Male 100 mm CL	284	0.04	1	0.840

Status within each year

Table C.2: Test of zone status within each survey year for each western rock lobster metric. The ratio is predicted lobster per pot in No-Take divided by Fished, with depth held at its mean, so a ratio above 1 indicates more lobster in No-Take.

Metric	Year	Ratio	SE	z	p
All lobster	2025	2.59	1.47	1.68	0.093
All lobster	2026	2.35	1.36	1.48	0.140
Sub-maturity (<75 mm CL)	2025	0.89	0.68	-0.15	0.881
Sub-maturity (<75 mm CL)	2026	1.04	0.80	0.05	0.957
Female 75 mm CL	2025	1.74	0.89	1.08	0.279
Female 75 mm CL	2026	1.74	0.95	1.02	0.306
Male 75 mm CL	2025	2.39	1.14	1.83	0.068
Male 75 mm CL	2026	3.30	1.73	2.27	0.023
Female 100 mm CL	2025	2.07	1.04	1.45	0.148
Female 100 mm CL	2026	2.72	3.47	0.79	0.432
Male 100 mm CL	2025	3.21	1.31	2.86	0.004
Male 100 mm CL	2026	2.89	1.46	2.10	0.036

Predicted values

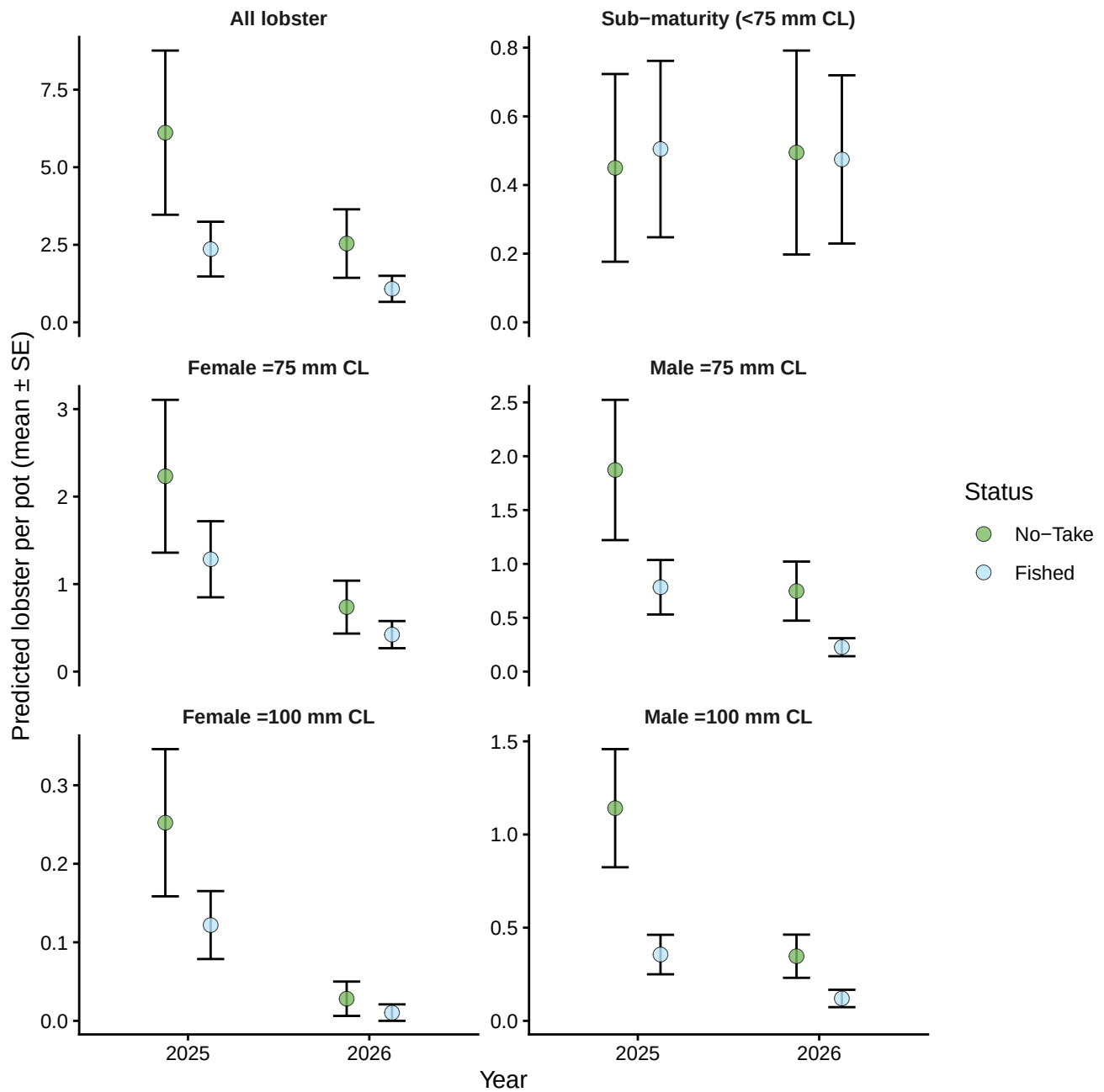


Figure C.1: Predicted western rock lobster per pot ($\pm$ SE) by zone status and survey year for each metric, from the negative binomial generalised linear mixed models, with depth held at its mean. Note each panel has its own y axis scale.

Table C.3: Predicted western rock lobster per pot by metric, zone status and survey year, with standard errors and asymptotic 95% confidence limits. Depth is held at its mean.

Metric	Status	Year	Per pot	SE	Lower CL	Upper CL
All lobster	Fished	2025	2.36	0.88	1.13	4.91
All lobster	No-Take	2025	6.11	2.65	2.62	14.28
All lobster	Fished	2026	1.08	0.42	0.50	2.31
All lobster	No-Take	2026	2.54	1.10	1.08	5.95
Sub-maturity (<75 mm CL)	Fished	2025	0.50	0.26	0.19	1.37
Sub-maturity (<75 mm CL)	No-Take	2025	0.45	0.27	0.14	1.48
Sub-maturity (<75 mm CL)	Fished	2026	0.47	0.25	0.17	1.31
Sub-maturity (<75 mm CL)	No-Take	2026	0.49	0.30	0.15	1.61
Female 75 mm CL	Fished	2025	1.28	0.43	0.66	2.49
Female 75 mm CL	No-Take	2025	2.23	0.87	1.04	4.81
Female 75 mm CL	Fished	2026	0.42	0.16	0.21	0.87
Female 75 mm CL	No-Take	2026	0.74	0.30	0.33	1.65
Male 75 mm CL	Fished	2025	0.78	0.25	0.42	1.48
Male 75 mm CL	No-Take	2025	1.87	0.65	0.95	3.70
Male 75 mm CL	Fished	2026	0.23	0.08	0.11	0.47
Male 75 mm CL	No-Take	2026	0.75	0.27	0.36	1.53
Female 100 mm CL	Fished	2025	0.12	0.04	0.06	0.24
Female 100 mm CL	No-Take	2025	0.25	0.09	0.12	0.52
Female 100 mm CL	Fished	2026	0.01	0.01	0.00	0.08
Female 100 mm CL	No-Take	2026	0.03	0.02	0.01	0.13
Male 100 mm CL	Fished	2025	0.36	0.11	0.20	0.64
Male 100 mm CL	No-Take	2025	1.14	0.32	0.66	1.97
Male 100 mm CL	Fished	2026	0.12	0.05	0.06	0.26
Male 100 mm CL	No-Take	2026	0.35	0.12	0.18	0.67

Model coefficients

Table C.4: Coefficients of the negative binomial generalised linear mixed models of western rock lobster per pot, on the log scale. Reference levels are Fished in the first survey year, so the status term is the effect of protection in that year and the interaction term is the change in that effect in the later year.

Metric	Term	Estimate	SE	z	p
All lobster	(Intercept)	0.859	0.374	2.29	0.022
All lobster	statusNo-Take	0.952	0.566	1.68	0.093
All lobster	year2026	-0.782	0.207	-3.78	<0.001
All lobster	depth_z	-0.039	0.188	-0.21	0.835
All lobster	statusNo-Take:year2026	-0.097	0.321	-0.30	0.763
Sub-maturity (<75 mm CL)	(Intercept)	-0.684	0.509	-1.34	0.179
Sub-maturity (<75 mm CL)	statusNo-Take	-0.115	0.767	-0.15	0.881
Sub-maturity (<75 mm CL)	year2026	-0.062	0.271	-0.23	0.821
Sub-maturity (<75 mm CL)	depth_z	-0.404	0.276	-1.47	0.143
Sub-maturity (<75 mm CL)	statusNo-Take:year2026	0.156	0.481	0.32	0.745
Female 75 mm CL	(Intercept)	0.250	0.339	0.74	0.461
Female 75 mm CL	statusNo-Take	0.553	0.511	1.08	0.279
Female 75 mm CL	year2026	-1.112	0.238	-4.67	<0.001
Female 75 mm CL	depth_z	0.125	0.202	0.62	0.536
Female 75 mm CL	statusNo-Take:year2026	0.003	0.370	0.01	0.993
Male 75 mm CL	(Intercept)	-0.244	0.323	-0.76	0.450
Male 75 mm CL	statusNo-Take	0.871	0.477	1.83	0.068
Male 75 mm CL	year2026	-1.241	0.259	-4.80	<0.001
Male 75 mm CL	depth_z	0.220	0.169	1.30	0.193
Male 75 mm CL	statusNo-Take:year2026	0.323	0.372	0.87	0.385
Female 100 mm CL	(Intercept)	-2.104	0.355	-5.93	<0.001
Female 100 mm CL	statusNo-Take	0.727	0.503	1.45	0.148
Female 100 mm CL	year2026	-2.467	1.084	-2.28	0.023
Female 100 mm CL	depth_z	0.605	0.265	2.28	0.023
Female 100 mm CL	statusNo-Take:year2026	0.274	1.366	0.20	0.841
Male 100 mm CL	(Intercept)	-1.033	0.298	-3.47	<0.001
Male 100 mm CL	statusNo-Take	1.166	0.408	2.86	0.004
Male 100 mm CL	year2026	-1.086	0.382	-2.85	0.004
Male 100 mm CL	depth_z	0.201	0.160	1.26	0.207
Male 100 mm CL	statusNo-Take:year2026	-0.105	0.520	-0.20	0.840